

Ahmed Ikram | Data Engineering

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Final-year BSc Computer Science (Data Science & AI) student at the University of Dundee (expected First-Class, 2027). Experienced in designing and building data platforms across three paradigms - Kafka event streaming on AWS, Databricks/dbt lakehouse ETL on Azure, and ClickHouse **OLAP** streaming on GCP - matching the architecture to the traffic pattern. A self-hosted ClickHouse pipeline ingested 58.9M+ events in 3 days with zero loss across **8 chaos injections**; the same quality-gate discipline covers model artifacts via Airflow retraining, MLflow promotion, and Evidently drift checks. Seeking a **Data Engineering** role serving BI and/or ML consumers.

EDUCATION

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- On track for a **First-Class Honours** degree. Expected Graduation: **June 2027**

TECHNICAL SKILLS

Languages & Databases: Python · SQL · T-SQL · PostgreSQL · Azure SQL (SQL Server) · Redis · SQLite · Alembic

Orchestration & Transformation: Apache Airflow · Apache Kafka · dbt · Apache Spark / PySpark · Pydantic · MLflow

Streaming & Warehousing: OpenLineage · Marquez · ClickHouse · Delta Lake · BigQuery · Databricks (DLT, Unity Catalog) · Azure SQL · Power BI

Data Quality & Observability: Great Expectations · Grafana · Prometheus · Azure Monitor · Cloud Monitoring · CloudWatch · StatsD · Evidently AI

Cloud & DevOps: AWS (ECS Fargate, EC2, S3, RDS, IAM, CloudFront, SageMaker, Secrets Manager) · Azure (Databricks, Azure SQL, ADLS) · Google Cloud (Compute Engine, BigQuery, GCS, Secret Manager, Workload Identity Federation) · Terraform · Docker · GitHub Actions · Linux

PROJECTS

W3C Web Logs ETL Pipeline

Airflow · Databricks · dbt · PySpark · Azure SQL · Delta Lake · Terraform · Power BI · Marquez · Grafana · Prometheus

- Databricks DLT medallion pipeline (Bronze → Silver) ingests 93 IIS log files (153,380 rows, 0 dropped) via 7 DLT quality gates; a consolidated MaxMind GeoIP struct UDF runs 3.5× faster than 7 separate UDFs and reaches 99.99% known-country coverage across 153,377 enriched rows.
- 16 dbt models build a Kimball-style star schema (9 SCD Type 1 dimensions; SCD Type-2 on GeoLocation) on hashed natural keys, compiling to PostgreSQL and T-SQL from one codebase via 18 compatibility macros; 121 dbt tests gate every model before Power BI and catch 213 duplicate rows on a 12-component incremental deduplication key.
- 2 Airflow DAGs orchestrate Databricks with Dataset-triggered decoupling of ingest from transform; OpenLineage → Marquez records the cross-engine graph (DLT → Azure SQL → dbt → Power BI) that Unity Catalog cannot see, including dbt test outcomes on the producing node; a 2-part Terraform estate (30+ Azure resources, OIDC federation, rollback workflow) deploys the platform.
- JDBC export of 153,377 Silver rows to Azure SQL dropped from 413s to 45s under serverless DLT's no-JDBC-write constraint; a freshness probe and 3 Grafana dashboards (23 panels, 10 alert rules) back a 7-page Power BI report (156K requests, 88 countries), with 627 tests across 5 tiers including dual-dialect dbt CI.

WikiStream Real-Time Streaming Analytics Platform

Python · ClickHouse · Grafana · BigQuery · Terraform · Google Cloud · Pydantic · Great Expectations

- Async consumer streams Wikipedia's EventStreams feed into a self-hosted ClickHouse cluster - 58.9M+ events in 3 days, 203.5 events/sec average, real peaks of 2,719/sec; Pydantic schema-on-write rejects malformed events to a dead-letter table before they hit disk.
- At-least-once delivery** via a durable cursor and a dedup ring buffer, proven with a live SIGKILL-mid-insert test (zero loss, zero duplicates); an 8-injection chaos battery fired and cleared every Grafana and Cloud Monitoring rule.
- 3 materialized views cut dashboard query latency 15x at p50 (13.2x p99) and rows scanned ~200x, holding 5,655 events/sec (2.08x real peak) with zero drops; hourly ClickHouse → GCS → BigQuery export keeps a 30-day-plus warehouse tier, parity-checked by exact-sum comparison every run.

- Great Expectations gates freshness, nulls, cardinality and schema drift hourly via systemd timers that also run backup, export and parity; a GCS restore replayed 4,514,837/4,514,837 rows exactly; GCP via Terraform + WIF, itemised **FinOps** tracking held to \$41.65/month run-rate, validated by 143 tests at 99.4% coverage.

ATM Log Aggregation, Anomaly Detection & Diagnostics Platform

Apache Kafka · FastAPI · PostgreSQL · Redis · MLflow · SageMaker · Docker · Terraform · AWS

- Data engineering lead on a 7-person Agile team for NCR Atleos; replaced a direct-DB generator with an event-driven **Kafka Raft** (KRaft) pipeline processing 2.5M+ events/day at 100+ msg/sec across 7 log sources, routing malformed records to a dead-letter table without halting ingestion.
- At-least-once delivery with idempotent dedup (Redis SET+TTL, LRU fallback) holds effectively-once behaviour across consumer restarts; Redis Stream DLQ with exponential backoff isolates poison messages before manual triage.
- PostgreSQL **data lake** on JSONB payloads - new sources only need a new parser, zero migrations; 14 indexes (6 composite) hold sub-100ms queries past 2M+ rows under a pooled, retrying connection layer, and feed a downstream detector with a SageMaker cross-check.
- MLflow (RDS backend + S3 artifacts) tracks every training run with champion-alias promotion, so ML consumers read a gated artifact; AWS estate of 10 Terraform modules / 118 resources (multi-AZ VPC, ECS Fargate, RDS, S3, CloudFront, 6 least-privilege IAM roles) with ingestion tests catching data loss under 50 concurrent writes.

StockLens Feature & Retraining Pipeline

Airflow · AWS · PyTorch · MLflow · Evidently AI · PostgreSQL · Redis · Terraform

- Weekly Airflow DAG runs 6-year OHLCV ingest → 17-feature build → MLflow-tracking training → champion/challenger promotion and blocks any ship that does not beat the live champion by >2pp so a regression cannot reach consumers.
- Evidently AI drift monitors PSI/KS/JSD on live vs training distributions beside the DAG and alerts to CloudWatch before drift reaches the champion - the same quality-gate pattern as the data pipelines, applied to a model artifact.
- Champion delivery writes the gated 'model.pt' to EFS (zero-copy inference) with S3 and a PostgreSQL model registry, so serving never reads an unpromoted run; prediction logs retain 90 days for offline replay.

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	<i>Feb 2025</i>
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	<i>Oct 2024</i>
<u>AWS Academy Graduate - Cloud Foundations</u>	<i>Mar 2024</i>

EXPERIENCE

Beaully Mini Market, Dundee, Scotland

Retail Assistant

September 2023 - Present

Zulus Peri Peri Grill, Dundee, Scotland

Front of House Worker

June 2021 - September 2023